

# Brightwater Analytics: Public API Reference v2

*Fictional sample document for demo purposes.*

## Overview

The Brightwater API gives programmatic access to datasets, reports and scheduled jobs. All endpoints are served over HTTPS from <https://api.brightwater.example/v2>. Requests and responses use JSON encoded in UTF-8.

## Authentication

Authenticate every request with an API key sent in the Authorization header using the Bearer scheme. API keys are created in the dashboard under Settings, then API Keys. Each key is scoped to one workspace. Keys can be rotated at any time; the old key keeps working for 24 hours after rotation so you can deploy the new one without downtime.

## Environments

Sandbox keys start with `bw_test_` and live keys start with `bw_live_`. Sandbox data is reset every Sunday at 02:00 UTC.

## Rate limits

The API allows 120 requests per minute per API key on the Standard plan and 600 requests per minute on the Enterprise plan. Report exports are additionally limited to 10 concurrent jobs per workspace. When a limit is exceeded the API returns HTTP 429 with a Retry-After header that states how many seconds to wait.

## Pagination

List endpoints are paginated with cursors. Pass the limit parameter (maximum 100, default 20) and use the next\_cursor value from the response to fetch the following page. A null next\_cursor means there are no more results.

## Idempotency

POST requests accept an Idempotency-Key header. If a request with the same key is received within 24 hours, the original response is returned instead of performing the action again.

## Endpoints

GET /datasets lists datasets in the workspace. POST /datasets creates a dataset from an uploaded CSV of up to 500 MB. GET /reports/{id} returns report metadata. POST /reports/{id}/export starts an asynchronous export and returns a job id. GET /jobs/{id} returns job status: queued, running, succeeded or failed.

## Errors

Errors use standard HTTP status codes and a JSON body with code and message fields. 400 means the request was invalid, 401 means the API key is missing or invalid, 403 means the key lacks permission for the workspace, 404 means the resource does not exist, and 5xx errors are safe to retry with exponential backoff.

## Webhooks

Webhooks notify you when an export job finishes. Each delivery is signed with HMAC-SHA256 in the Brightwater-Signature header. Failed deliveries are retried up to 8 times over 24 hours.